

Gavin A Hayes

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Skills

Systems Programming: Cross-platform and embedded development in C and C++; build systems (Make, CMake); CI/CD

WebAssembly: Porting existing software (Emscripten, wasi-sdk); plugin systems and integration (Extism with XTP bindings generation); tooling (binary parsing/recompilation with Binaryen and wasmparser)

Backend Development: Rust, Typescript/Node, Perl, PostgreSQL, Fastify, CI/CD

Technical Writing: Blog posts, tutorials, reference material, conference presentations

Experience

Keycard Labs - *Member of Technical Staff (Remote)*

December 2025 - June 2026, contractor September 2025 - November 2025

- Built an OAuth compatible MCP gateway in Rust using the [Pingora](#) framework. Enables agents to connect to third-party OAuth MCP servers without exposing upstream tokens via token exchange.
- Added [email & password and social logins](#) to a customer configurable login system including supporting register with email verification and login with password reset within a MCP server connect flow.

Dylibso - *Senior Software Engineer (Remote)*

May 2023 - August 2025

- Built an instrumenting recompiler for adding observability to existing WebAssembly modules using [Binaryen](#). Co-hosted [Wasm Observability Workshop](#).
- Developed the cross-language [Extism framework](#) for building with WebAssembly. Worked on C++, Python, C, and Perl support. Presented [Extism Perl SDK at The Perl & Raku Conference 2024](#).
- Developed XTP Processor, a queue and processing system (Rust, PostgreSQL, Typescript) to test and validate uploaded WebAssembly modules conformed to an interface.
- Added [C++ support](#) and improved [Python support](#) to OpenAPI-like schema bindings generator (XTP bindgen) to ease WebAssembly plugin development.
- Built full-stack OAuth client and server implementations for [mcp.run](#) to enable secure integration between AI applications and external tools/services via the Model Context Protocol (MCP). Performed production operations including deployments, database migration troubleshooting and manual fixes, log monitoring, and system observability with Datadog.

TestVonics - *a Raptor Scientific Company - Lead Software Engineer (Peterborough, NH)*

May 2017 - May 2023, interned summers 2014-2016 and winter 2016

- Built a touchscreen handheld remote for an Air Data Test Set using C, C++, Qt, and Debian linux on a BeagleBone.
- Upgraded and maintained a Win32 MFC based Air Data Test Set and Calibrator, including migration from VC++6 and Windows XP to VS2017 and Windows 10 IoT Enterprise, as well as the addition of Quick Leak Test, STEP function, and safety features and QoL improvements.

Open Source Software Development

- Actually Portable Perl (2022 - PRESENT) - Ported Perl to the Cosmopolitan Libc. Created single-file self-contained builds of Perl that run on six operating systems with the same binary. Created a toolkit for creating binary releases of Perl applications. computoid.com/posts/Perl-is-Actually-Portable.html computoid.com/APPerl/ [Talk at TPRC 2024](#)
- Cosmopolitan Libc (2022 - PRESENT) - Contributor and Maintainer, added directory support to the built-in ZIP file VFS, fixes to Windows emulation of POSIX capabilities. github.com/jart/cosmopolitan/pulls?q=is:pr+author:G4Vi

- Media HTTP File Server (2018 - PRESENT) - Built an event based HTTP media server and web app for streaming media via the browser or other applications using Perl, C, Javascript, and Wasm. github.com/G4Vi/MHFS
- psx_screen_dumper / screen_data_reader (2021-2022) - Built a pair of applications to exfiltrate the BIOS or save game data from a PlayStation via video by using OpenCV via Python. github.com/G4Vi/psx_screen_dumper github.com/G4Vi/screen_data_reader/
- mkpsxiso (2021-2022) - Improve CDDA pregap handling for bit-perfect unpacking and repacking of CD images. Add extracting CDDA as FLAC (instead of raw PCM) with repacking support to save disk space. github.com/Lameguy64/mkpsxiso/commits/master/?author=G4Vi
- tonyhax (2021) - Found a buffer overflow and developed MIPS shellcode to exploit a save game feature in a PlayStation game. github.com/socram8888/tonyhax/pull/62
- GIF Encoding (2021) - Added transparency support to a GIF encoder and ported the GIF encoder to Perl. github.com/lecram/gifenc/pulls?q=is%3Apr+author%3AG4Vi, github.com/G4Vi/gifenc, metacpan.org/pod/Image::GIF::Encoder::PP
- dr_flac (2020-2022) - Fixed infinite loop bug and removed unaligned 32-bit loads from metadata parsing in popular FLAC decoding library. github.com/mackron/dr_libs/pulls?q=is:pr+author:G4Vi

Formal Education

Worcester Polytechnic Institute, Worcester, MA - BS in Computer Science

August 2013 - May 2017

- Collaborated to make a new version of Bomblab in Python and C, a computer based x86 ASM reverse engineering assignment for senior year MQP project.